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Step 1: Import Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from scipy import stats
import statsmodels.api as sm
from statsmodels.formula.api import ols
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
```

Step 2: Load Dataset

Download IBM HR Employee Attrition Dataset and upload it to Colab.

```
df = pd.read_csv("/content/IBM-HR-Analytics-Employee-Attrition-and-Performance-Revised.csv")
df.head()
```

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EnvironmentSatisfaction
0	41	Yes	Travel_Rarely	1102	Sales	1	College	Life Sciences	Medium
1	49	No	Travel_Frequently	279	Research & Development	8	Below College	Life Sciences	High
2	37	Yes	Travel_Rarely	1373	Research & Development	2	College	Other	Very High
3	33	No	Travel_Frequently	1392	Research & Development	3	Master	Life Sciences	Very High
4	27	No	Travel_Rarely	591	Research & Development	2	Below College	Medical	Low

5 rows × 31 columns

Step 3: Descriptive Statistics

Mean, Median, Mode

```
income = df["MonthlyIncome"]

print("Mean:", income.mean())
print("Median:", income.median())
print("Mode:", income.mode()[0])
```

```
Mean: 6502.931292517007
Median: 4919.0
Mode: 2342
```

Variance and Standard Deviation

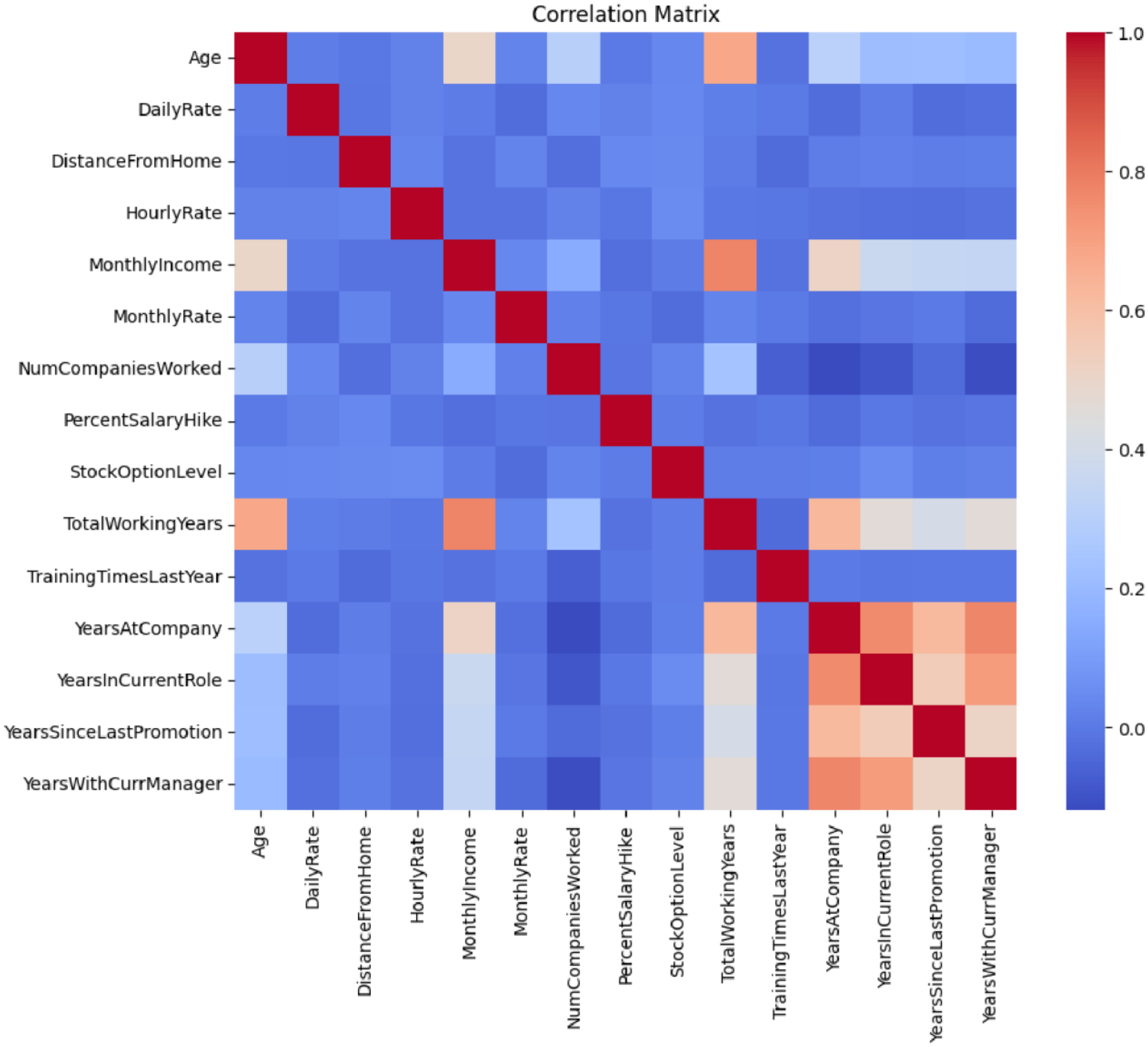
```
print("Variance:", income.var())
print("Standard Deviation:", income.std())
```

```
Variance: 22164857.07151842
Standard Deviation: 4707.956783097995
```

Step 4: Pearson Correlation Analysis Select

numerical columns.

```
numeric_df = df.select_dtypes(include=np.number)

correlation = numeric_df.corr()

plt.figure(figsize=(10,8))
sns.heatmap(correlation, cmap="coolwarm")
plt.title("Correlation Matrix")
plt.show()
```



Example Correlation

```
print(df["Age"].corr(df["MonthlyIncome"]))
```

```
0.4978545669265805
```

Step 5: Hypothesis Formulation Business Problem

Is there a significant difference in monthly income between employees who leave the company and those who stay?

Null Hypothesis (H_0)

There is no significant difference in Monthly Income between employees who leave and those who stay.

Alternative Hypothesis (H_1)

There is a significant difference in Monthly Income between the two groups.

Step 6: Independent Sample t-test

```
leave = df[df["Attrition"]=="Yes"]["MonthlyIncome"]
stay = df[df["Attrition"]=="No"]["MonthlyIncome"]

t_stat, p_value = stats.ttest_ind(leave, stay)

print("T-statistic:", t_stat)
print("P-value:", p_value)
```

```
T-statistic: -6.203935765608938
P-value: 7.147363985353811e-10
```

Decision

```
if p_value < 0.05:
    print("Reject Null Hypothesis")
else:
    print("Fail to Reject Null Hypothesis")
```

```
Reject Null Hypothesis
```

Step 7: One-Way ANOVA

Compare Monthly Income across different Job Roles.

```
model = ols('MonthlyIncome ~ C(JobRole)', data=df).fit()
anova = sm.stats.anova_lm(model, typ=2)

print(anova)
```

```
                  sum_sq      df           F  PR(>F)
C(JobRole)  2.657098e+10     8.0  810.214054     0.0
Residual    5.989190e+09  1461.0         NaN     NaN
```

Step 8: Linear Regression

Analyze the relationship between YearsAtCompany and MonthlyIncome.

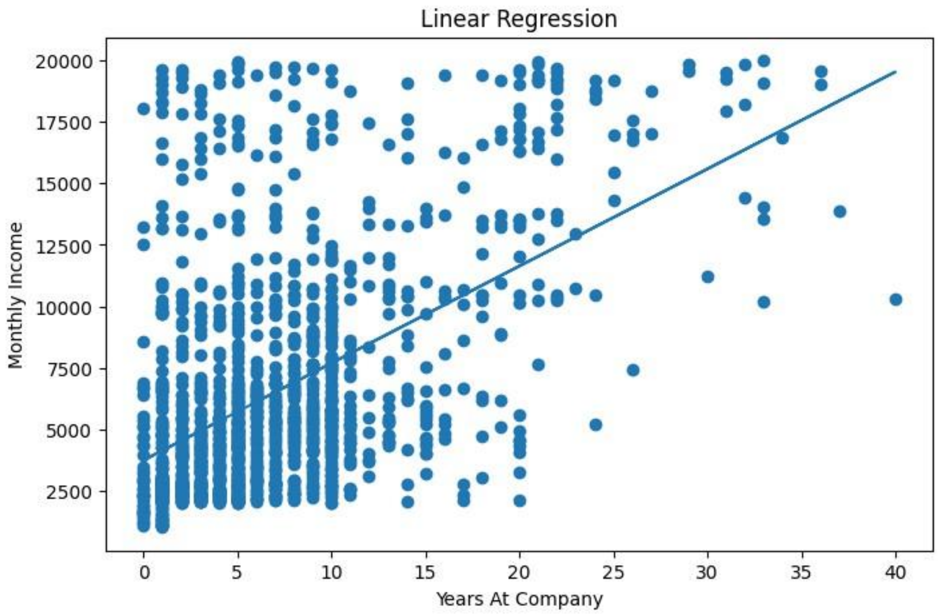
```
X = df[["YearsAtCompany"]]
y = df["MonthlyIncome"]

model = LinearRegression()
model.fit(X, y)

prediction = model.predict(X)

print("Intercept:", model.intercept_)
print("Slope:", model.coef_[0])
print("R² Score:", r2_score(y, prediction))
```

```
Intercept: 3733.2731481323835
Slope: 395.20456923368243
R² Score: 0.26448888197942455
```

Regression Graph

```
plt.figure(figsize=(8,5))
plt.scatter(X, y)
plt.plot(X, prediction)
plt.xlabel("Years At Company")
```

```
plt.xlabel("Years At Company")
plt.ylabel("Monthly Income")
plt.title("Linear Regression")
plt.show()
```



Step 9: Interpretation of Statistical Outputs

Output Meaning p-value Shows statistical significance Confidence Interval Range where the true value is likely to lie Regression Coefficient Change in income for each additional year R^2 Score Percentage of variance explained by the model Example p = 0.02 → Significant.

$R^2 = 0.45$ → Model explains 45% of income variation.

Step 10: Conclusion

The analysis shows that descriptive statistics summarize employee income, correlation identifies relationships among variables, hypothesis testing validates business assumptions, and linear regression predicts salary trends based on years at the company. These techniques help organizations make informed HR decisions.